

Microsoft Entra ID IAM Lifecycle Management Project Report

Microsoft Entra ID Identity and Access Management (IAM) Lifecycle Project

Project Name: Microsoft Entra ID IAM Lifecycle Management

Technology: Microsoft Entra ID (Azure Active Directory)

Category: Identity and Access Management (IAM)

Platform: Microsoft Entra ID Free Tenant

Project Type: Hands-on Enterprise IAM Lab

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1. Introduction

Identity and Access Management (IAM) is a critical component of enterprise security. Organizations use Microsoft Entra ID to securely manage user identities, authentication, authorization, and access to organizational resources.

This project demonstrates the implementation of core IAM operations within a Microsoft Entra ID Free tenant. The lab simulates real-world identity lifecycle activities performed by IAM administrators, including onboarding new employees, modifying access when employees change departments, and securely removing access when employees leave the organization.

The project also covers administrative role assignments, security group management, Multi-Factor Authentication (MFA), and monitoring user sign-in activity through audit logs.

2. Objectives

The objectives of this project are:

- Deploy and configure a Microsoft Entra ID tenant.
 - Create and manage user accounts.
 - Organize users into Security Groups.
 - Assign administrative roles using the principle of least privilege.
 - Simulate Joiner, Mover, and Leaver (JML) lifecycle processes.
 - Configure Multi-Factor Authentication (MFA).
 - Monitor authentication events through Sign-in Logs.
 - Document IAM activities using screenshots and reports.
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3. Environment Setup

Platform

Microsoft Entra ID Free

Administrative Portal

Microsoft Entra Admin Center

Tenant

Sample Enterprise Tenant

Operating System

Windows 11

Browser

Microsoft Edge / Google Chrome

Tools Used

- Microsoft Entra ID
 - Microsoft 365 Admin Center
 - Security Groups
 - Microsoft Authenticator
 - GitHub
 - Microsoft Word
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4. Creating Users

A simulated enterprise environment was created by adding multiple employees representing different business departments.

Sample Users

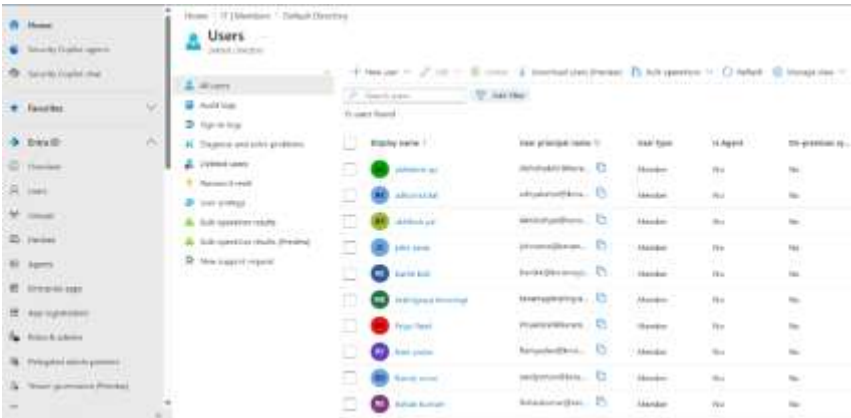
Display Name Department

Abhishek ap	Developers
Aditya Kattal	Finance
Arun Keramagi	Management
Akhilesh pal	Sales
Priya Patel	Human Resources
Rohan Kumar	Information Technology
Sumit Tiwari	Information Technology
Kartik Koli	Management

Each user was created with:

- User Principal Name (UPN)
- Display Name
- Temporary Password
- Department Assignment

Screenshot:



5. Creating Security Groups

Security Groups were created to simplify permission management.

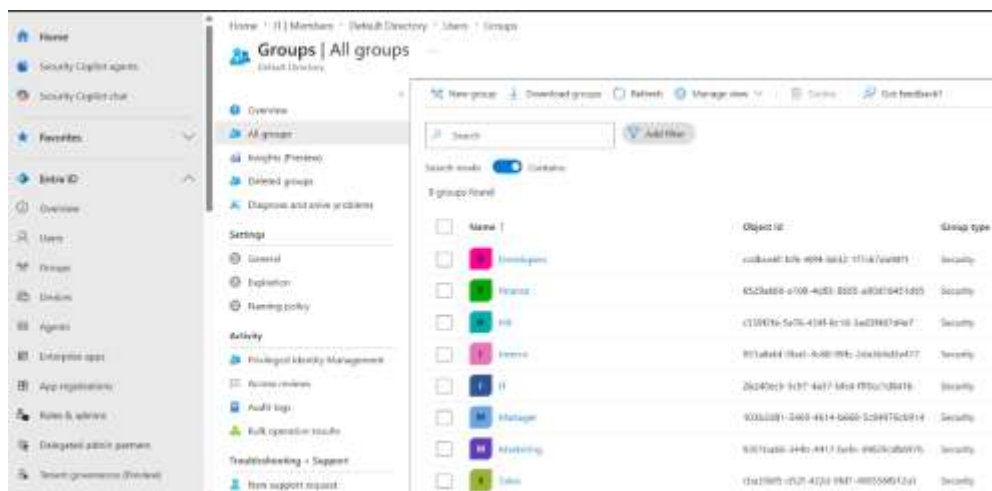
Groups Created

- HR
- IT
- Finance
- Marketing
- Sales
- Developers
- Managers

- Interns

Each user was assigned to the appropriate department group.

Screenshot:



6. Administrative Roles

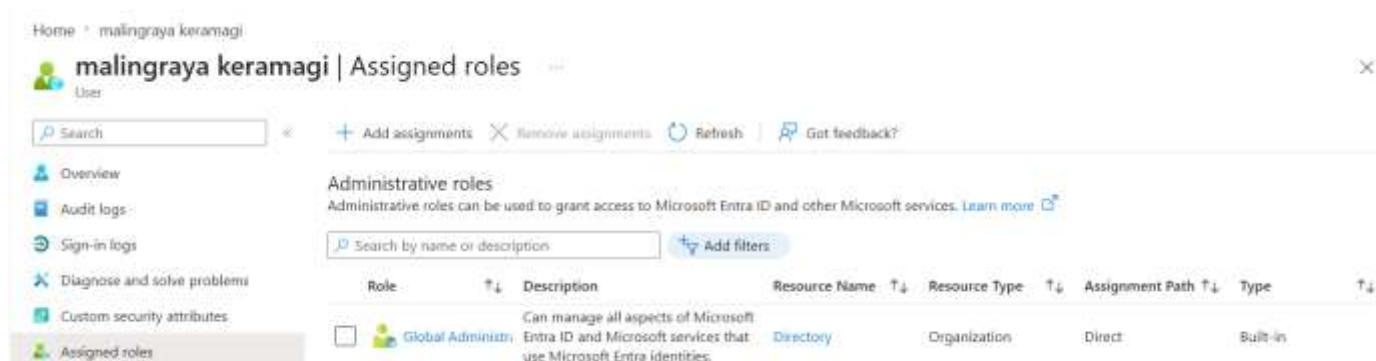
Administrative roles were assigned following the Principle of Least Privilege.

Roles Used

- Global Administrator
- User Administrator
- Groups Administrator
- Password Administrator

Administrative permissions were limited to only the responsibilities required for each role.

Screenshot:



7. Joiner Process

The Joiner process simulates onboarding a new employee.

Activities Performed

- Created a new user account.
- Assigned a temporary password.
- Added the employee to the IT Security Group.

- Assigned the appropriate administrative permissions where applicable.
- Enabled password reset during the first sign-in.

Outcome:

The employee received appropriate access based on organizational policies while maintaining security standards.

8. Mover Process

The Mover process demonstrates modifying user access after a departmental transfer.

Scenario

Employee transferred from Finance to IT.

Actions

- Removed Finance Security Group membership.
- Added IT Security Group membership.
- Verified updated permissions.
- Confirmed access reflected new job responsibilities.

Outcome:

User access was successfully updated without creating a new account.

9. Leaver Process

The Leaver process demonstrates secure user offboarding.

Activities

- Blocked user sign-in.
- Reset account password.
- Removed group memberships.
- Revoked organizational access.
- Deleted the user account.

Outcome:

Former employee access was completely removed from the tenant.

10. Multi-Factor Authentication (MFA)

To strengthen identity security, Multi-Factor Authentication was configured for privileged accounts.

Configuration

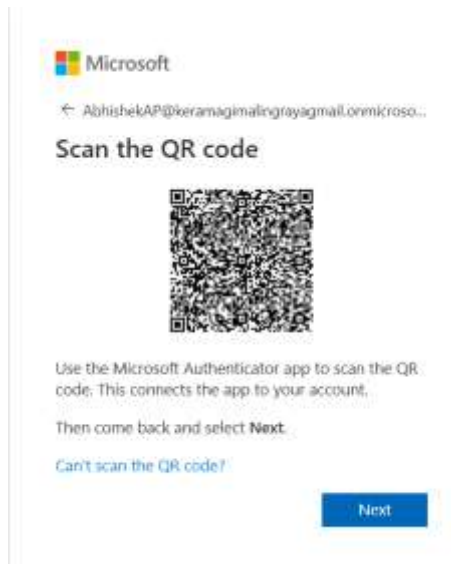
- Enabled MFA.
- Registered Microsoft Authenticator.
- Tested successful authentication.

- Verified second-factor verification.

Benefits

- Protection against password compromise.
- Enhanced identity verification.
- Improved account security.

Screenshot:



11. Sign-in Logs

Authentication activity was monitored through Microsoft Entra Sign-in Logs.

Events Reviewed

- Successful Login
- Failed Login
- Password Reset
- MFA Authentication
- User Activity

Monitoring authentication logs helps administrators detect suspicious activity and troubleshoot user access issues.

Screenshot:



12. Security Best Practices

The following best practices were applied throughout the project:

- Principle of Least Privilege
 - Role-Based Access Control (RBAC)
 - Security Group Management
 - Strong Authentication using MFA
 - User Lifecycle Management
 - Regular Audit Log Monitoring
 - Controlled Administrative Access
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13. Learning Outcomes

This project provided practical experience in:

- Identity Lifecycle Management
 - Microsoft Entra ID Administration
 - User and Group Management
 - Administrative Role Assignment
 - RBAC Concepts
 - Security Group Administration
 - MFA Configuration
 - Identity Governance Fundamentals
 - Enterprise IAM Operations
 - Security Monitoring
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14. Conclusion

This project successfully demonstrated enterprise Identity and Access Management operations using Microsoft Entra ID. Core IAM activities—including user provisioning, security group management, role assignments, Joiner-Mover-Leaver lifecycle processes, Multi-Factor Authentication, and sign-in log monitoring—were implemented in a controlled lab environment.

The project reflects common responsibilities of IAM Administrators and provides practical experience aligned with real-world enterprise identity management practices. The documentation, screenshots, and GitHub repository together form a professional portfolio project suitable for internship and entry-level IAM roles.